

GEN ACADEMY

Week 3 Project

SprintPulse GTM Content Agent: Ideation to Copy

A Low-Code Multi-Agent Workflow Using n8n, Google Sheets, and Groq

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1. Project Overview

SprintPulse GTM Content Agent is a low-code, multi-agent workflow that transforms a natural-language campaign request into a coordinated set of go-to-market content. The solution uses n8n to orchestrate specialized Strategy, Writer, Reviewer, and Revision agents.

The workflow validates the request, retrieves verified product information from a Google Sheets catalog, creates a grounded content strategy, generates four marketing formats, independently reviews the draft, and permits no more than one automated revision. The final content is marked as pending human review and is never published automatically.

The prototype supports three fictional products: SprintPulse AI, LaunchPilot, and SupportSense AI. All product information and evaluation data are synthetic.

2. Problem Statement

Marketing teams often need to convert limited product information into multiple coordinated campaign assets. Completing this work manually can require separate planning, writing, fact-checking, revision, and approval steps. It can also produce inconsistent messaging or unsupported claims when product facts are not centrally controlled.

This project addresses that problem with a multi-agent workflow. Each agent has a defined responsibility, while verified product data from the catalog acts as the source of truth. The workflow generates content efficiently while preserving grounding, review, error handling, and human approval controls.

3. Goal and Success Criteria

The goal was to build a low-code GTM content agent that could accept a simple campaign request and produce a grounded campaign package containing a LinkedIn post, promotional email, short blog, and exactly three advertisement variations.

The success criteria were:

- Correctly identify supported products from natural-language input.
- Retrieve only the corresponding product's verified catalog record.
- Detect the requested tone or apply professional as the default.
- Generate all four required content formats.
- Prevent unsupported facts, invented capabilities, and unverified performance claims.
- Use an independent reviewer before returning the content.
- Permit no more than one automated revision.
- Route missing, unknown, malformed, or failed outputs to clear error responses.
- Finish with a pending-human-review status.
- Never publish content automatically.
- Successfully handle at least 8 of 10 evaluation scenarios after targeted improvements.

4. Use Case and Target User

The primary use case is generating a coordinated go-to-market campaign for a product stored in the verified catalog. A user submits a request such as “Create a professional campaign for SprintPulse AI.” The workflow identifies the product and tone, retrieves the matching catalog record, and generates a campaign package for human review.

Target users include product marketers, startup founders, campaign managers, and small marketing teams that need an initial set of coordinated content assets while maintaining control over product claims and final approval.

Typical user flow:

- 1. The user submits a campaign request through the n8n chat interface.
- 2. The workflow extracts the product name, requested tone, and optional URL.
- 3. The request is validated.
- 4. The matching product is retrieved from the catalog.
- 5. The Strategy Agent prepares grounded messaging guidance.
- 6. The Writer Agent creates the required campaign assets.
- 7. The Reviewer Agent evaluates grounding, completeness, and tone.
- 8. If necessary, the Revision Agent performs one correction.
- 9. The completed package is returned with a pending-human-review status.

5. Dataset and Privacy

The project uses a synthetic Google Sheets product catalog containing three fictional products: SprintPulse AI, LaunchPilot, and SupportSense AI. Each catalog record includes the product name, description, key features, pricing, target audience, launch date, and category.

No real customer, employee, company, or confidential business data is used. The repository contains an exported copy of the synthetic catalog so reviewers can inspect or reproduce the project.

API keys, passwords, personal email addresses, credential identifiers, and the original Google Sheet document ID are excluded from the repository. The exported n8n workflow uses placeholders and requires reviewers to configure their own credentials and product-catalog reference after import.

Product Name	Description	Key Features	Pricing	Target Audience	Launch Date	Category
SprintPulse AI	SprintPulse AI is a project-status assistant that turns team updates into concise weekly reports and highlights delivery risks for human review.	Weekly status summaries, blocker detection, overdue-task alerts, risk flags, leadership-ready reports	Starter: \$19 per user per month. Team: \$49 per user per month with shared reporting and integrations.	Project managers, program managers, engineering managers, and delivery leads	October 15, 2026	Project Management SaaS
LaunchPilot	LaunchPilot is a campaign-planning workspace that helps small marketing teams organize launch tasks, messaging, channel plans, and approval deadlines.	Campaign checklists, messaging planner, channel calendar, approval tracking, launch-readiness dashboard	Starter: \$29 per month for up to 3 users. Growth: \$79 per month for up to 10 users.	Startup founders, product marketers, and small marketing teams	November 10, 2026	Marketing SaaS
SupportSense AI	SupportSense AI analyzes customer-support conversations and organizes recurring issues into themes that support teams can review and prioritize.	Conversation categorization, recurring-issue detection, sentiment summaries, escalation flags, weekly insight reports	Starter: \$39 per month for 1 support workspace. Business: \$99 per month with multiple workspaces and advanced reporting.	Customer-support managers, support operations teams, and customer-success leaders	January 20, 2027	Customer Support SaaS

Figure 1. Synthetic product catalog used as the verified source of truth.

6. System Architecture

The solution uses three main platforms:

- n8n Cloud orchestrates the workflow, hosts the chat interface, validates data, routes decisions, and formats the final response.
- Google Sheets acts as the verified product catalog and source of truth.
- Groq-hosted language models power the Strategy, Writer, Reviewer, revision, and structured-output repair steps.

The architecture separates deterministic workflow logic from generative AI tasks. n8n handles validation, catalog matching, branching, parsing, retry protection, and final status control. The language models handle strategy development, content generation, independent review, and one permitted revision.

Architecture flow:

User request → Input normalization → Validation → Google Sheets catalog lookup → Product matching → Strategy Agent → Writer Agent → Structured-output validation → Reviewer Agent → PASS or REVISE decision → Optional single revision → Human-review response

No branch connects to a publishing platform, ensuring that generated content cannot be published automatically.

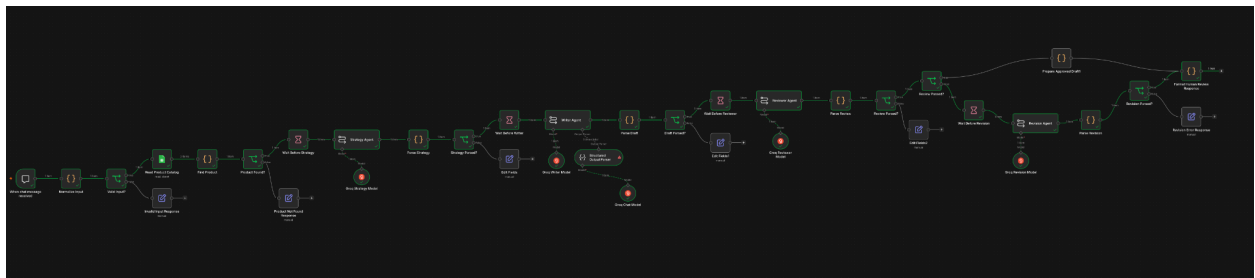


Figure 2. Complete SprintPulse GTM multi-agent workflow in n8n.

7. End-to-End Workflow

The workflow begins when the n8n chat trigger receives a user request. The Normalize Input node extracts the requested product, tone, and optional URL. The Valid Input decision routes incomplete requests to an explanatory response without making unnecessary AI calls.

Valid requests read the product catalog from Google Sheets. The Find Product node performs case-insensitive matching, and unknown products are routed to a Product Not Found response. For a supported product, the workflow prepares a verified product-data object and sends it to the Strategy Agent.

The resulting strategy is parsed and validated before being sent to the Writer Agent. The Writer produces a LinkedIn post, promotional email, short blog, and three advertisement variations. A Structured Output Parser checks the required JSON schema, while an auto-fix model can repair formatting when necessary.

The Reviewer Agent independently evaluates the original draft for grounding, completeness, and tone. A PASS verdict sends the draft directly to final preparation. A REVISE verdict permits one Revision Agent call. The workflow does not create an unlimited revision loop.

Both successful routes converge at the final formatter, which clearly states that the content is pending human review and has not been published. Parsing failures and invalid outputs are routed to dedicated error responses.

8. Agent Roles

The workflow uses specialized agents with separate responsibilities:

Strategy Agent:

Creates the campaign foundation before content is written. It summarizes the verified product, identifies the target audience and pain points, develops a value proposition, selects proof points, proposes messaging angles, establishes tone guidelines, and generates content hooks.

Writer Agent:

Uses only the verified product record and approved strategy to create four coordinated assets: one LinkedIn post, one promotional email, one short blog, and exactly three advertisement variations. It must follow the requested tone and the defined length and formatting rules.

Reviewer Agent:

Independently evaluates the original draft against the verified product data. It returns a PASS or REVISE verdict, grounding score, completeness score, tone score, identified issues, revision instructions, and a review summary. It does not rewrite the content.

Revision Agent:

Runs only when the Reviewer returns REVISE. It applies the review instructions once, while preserving the verified product facts, requested tone, required formats, and JSON structure. The revised content is not automatically approved or published.

Human Reviewer:

Makes the final approval decision outside the automated workflow. The final response remains pending human review regardless of whether an automated revision occurred.

11. Writer Agent

The Writer Agent receives the verified product record, requested tone, and parsed campaign strategy. It generates one LinkedIn post, one promotional email, one short blog, and exactly three advertisement variations.

The Writer uses the Groq-hosted openai/gpt-oss-20b model with temperature 0.1 and a maximum output allowance of 3,000 tokens. Its prompt defines content-length targets, required fields, grounding rules, tone requirements, JSON formatting, and the prohibition against unsupported measurable claims.

A Structured Output Parser validates the Writer response against the required schema. If the first response does not match that schema, an auto-fix model attempts to repair the structure. The repair model uses openai/gpt-oss-20b with temperature 0. The Parse Draft node supports both parsed object output and text-based JSON as a fallback.

This structured-output layer was introduced after baseline tests exposed truncated, malformed, and reasoning-contaminated JSON responses.

12. Reviewer and Revision Agents

The Reviewer Agent acts independently from the Writer. It compares the original campaign against the verified product data and requested tone. It returns a PASS or REVISE verdict together with grounding, completeness, and tone scores, specific issues, revision instructions, and a summary.

The Reviewer uses the Groq-hosted openai/gpt-oss-120b model with temperature 0.1 and a maximum output allowance of 4,000 tokens. Its response is parsed and validated before the workflow evaluates the verdict.

A PASS verdict sends the original campaign to final preparation. A REVISE verdict triggers one Revision Agent call. The Revision Agent uses openai/gpt-oss-20b with temperature 0.2 and a maximum output allowance of 3,000 tokens. A 30-second delay before revision reduces provider rate-limit failures.

The revision path is deliberately limited to one attempt. Revised output is not sent through a second automated review; instead, it is clearly marked as pending human review. Therefore, automated reviewer scores shown in the final result describe the original draft, not the revised campaign.

13. Human Review and Publishing Controls

The workflow is designed to produce a campaign draft, not to publish content. Both the PASS and REVISE routes end at the Format Human Review Response node with the status `pending_human_review`.

The final response explicitly states whether an automated revision was applied and confirms that nothing was published automatically. There are no connections to social-media, email-marketing, advertising, content-management, or other publishing platforms.

A human reviewer must verify product claims, tone, formatting, links, pricing, launch dates, and calls to action before using the generated content. This control prevents an AI-generated draft from becoming public without accountable approval.

```
"revision_applied": true,
"revision_parse_success": true,
"revision_parse_error": "",
"final_status": "pending_human_review",
"output": "# SprintPulse AI GTM Campaign\n\n> **Status: Pending human\n          review**\n\n> One automated revision was applied after the independent\n          review.\n\n> Nothing has been published automatically.\n\n## LinkedIn\nPost\n\nStruggling to keep weekly status reports concise? SprintPulse AI\nautomatically compiles team updates, flags blockers, and highlights\ndelivery risks. Deliver clear, risk-aware updates to stakeholders.\n\n#ProjectManagement #LeadershipReady\n\n## Promotional\nEmail\n\n**Subject:** Turn Team Updates into Leadership-Ready
```

Figure 4. Final campaign output showing successful revision and pending-human-review status.

14. Guardrails and Error Handling

The workflow includes the following guardrails:

- Use only the verified product record supplied by the catalog.
- Do not invent features, integrations, customer evidence, performance results, or measurable benefits.
- Do not claim that an optional URL was researched or verified.
- Apply the requested tone or use professional as the default.
- Generate all required content formats and exactly three ad variations.
- Separate strategy, writing, review, and revision responsibilities.
- Permit no more than one automated revision.
- Mark every successful result as pending human review.
- Never publish generated content automatically.
- Return controlled errors rather than guessing when required information is missing.

Dedicated error routes handle invalid input, unsupported products, strategy parsing failures, draft parsing failures, review parsing failures, and revision parsing failures. Wait nodes were added between larger model calls to reduce provider rate-limit failures.

Structured validation is applied to the Writer because evaluation showed that long, nested campaign responses were particularly likely to contain malformed JSON. The auto-fix model repairs formatting only; it does not replace human approval.

15. Evaluation Method

The workflow was evaluated with 10 test scenarios covering normal operation and edge cases:

1. Primary product with an explicit professional tone.
2. Second supported product with a conversational tone.
3. Third supported product with a friendly tone.
4. Missing product name.
5. Unknown product.
6. Supported product with no tone, requiring the professional default.
7. Supported product with an optional URL.
8. Case-insensitive product name with a custom tone.
9. An alternate request verb using “write” instead of “create.”
10. Minimal valid input containing only a supported product name.

For each test, the evaluation workbook records the user input, expected behavior, actual route, identified product and tone, Reviewer verdict, whether revision occurred, final status, grounding score, completeness score, latency, outcome, and notes.

The baseline execution was preserved without modification. After targeted workflow improvements, the three originally failing cases were rerun in a separate Retests After Improvements section. This approach maintains a transparent record of failures, changes, and subsequent results.

Automated grounding and completeness scores apply to the original draft evaluated by the Reviewer. When a revision occurs, the final revised campaign is manually inspected but is not assigned an invented automated score.

16. Evaluation Results

The baseline evaluation passed 7 of 10 scenarios, producing a 70% initial pass rate. The three failures were caused by intermittent empty model output or invalid and truncated JSON.

Targeted improvements included increasing model token allowances where necessary, reducing Writer output length, enforcing a structured Writer schema, adding automatic formatting repair, updating Parse Draft to accept object output, and adding delays and lower token limits to reduce rate-limit failures.

All three originally failing scenarios subsequently completed successfully in targeted retests. Based on the latest successful result for each scenario, functional coverage reached 10 of 10 use cases, exceeding the project target of at least 8 of 10.

Some successful runs followed the REVISE route and retained low Reviewer scores because those scores evaluate the original draft. The corrected campaign was returned with pending_human_review status and manually checked for required formats and verified product information.

Structured-output repair and rate-limit delays increased latency. Selected successful retests took approximately 56 to 70 seconds, demonstrating a tradeoff between output reliability and response speed.

Test #	Category	User Input	Expected Behavior	Actual Route	Product Identified	Tone Identified	Reviewer Verdict	Revision Applied	Final Status	Grounding Score	Com
1	Happy path — primary product	Create a professional campaign for SprintPulse AI	Find SprintPulse AI, use professional tone, generate all four formats, perform review, allow no more than one revision, and finish pending human review.	Draft error path	SprintPulse AI	professional	N/A	No	draft_error	N/A	
2	Happy path — second product	Create a conversational campaign for LaunchPilot	Find LaunchPilot, use conversational tone, use only LaunchPilot facts and pricing, generate all four formats, and finish pending human review.	REVISE path	LaunchPilot	conversational	REVISE	Yes	pending_human_review	30	
3	Happy path — third product	Create a friendly campaign for SupportSense AI	Find SupportSense AI, use friendly tone, use only SupportSense AI facts and pricing, generate all four formats, and finish pending human review.	REVISE path	SupportSense AI	friendly	REVISE	Yes	pending_human_review	30	
4	Missing product	Create a professional campaign for	Route to Invalid Input Response, request a product name, and make no AI model calls.	Invalid input path	Not identified	professional	N/A	No	invalid_input	N/A	
5	Unknown product	Create a professional campaign for TimeTracker Pro	Route to Product Not Found Response, list the available products, and make no AI model calls.	Product not found path	TimeTracker Pro — not found	professional	N/A	No	product_not_found	N/A	
6	Default tone	Create a campaign for LaunchPilot	Find LaunchPilot, assign professional as the default tone, generate all four formats, and finish pending human review.	Strategy parse error path	LaunchPilot	professional	N/A	No	strategy_error	N/A	
7	Optional URL	Create a professional campaign for SprintPulse AI https://example.com	Extract SprintPulse AI and the URL separately, but do not claim the URL was researched or verified.	REVISE path	SprintPulse AI	professional	REVISE	Yes	pending_human_review	30	

Evaluation Summary					
Metric	Result				
Baseline tests passed	7/10				
Baseline pass rate	70%				
Targeted retests passed	3/3				
Targeted retest pass rate	100%				
Latest functional coverage	10/10				
Latest functional success rate	100%				
Original-draft automated scores	Retained without modification				
Revised outputs automatically re-scored	No				
Human review required	Yes				
Automatic publishing enabled	No				
After the baseline evaluation, targeted improvements were applied to model token settings, structured Writer output, object-based draft parsing, and rate-limit handling. All three originally failing use cases subsequently completed successfully in targeted retests. Automated reviewer scores describe the pre-revision drafts because revised outputs are not submitted for a second automated review. Therefore, final revised campaigns remain pending human review and are not published automatically.					

Figure 5: Baseline and targeted-retest and final evaluation results

17. Failures and Iterations

The evaluation revealed several important failure modes and led to the following iterations:

Empty model responses

The Strategy Agent occasionally returned an empty text value, causing “Unexpected end of JSON input.” Parse validation correctly detected the failure and routed it to the Strategy Error Response. These intermittent provider failures were preserved in the baseline results.

Truncated or malformed Writer JSON

Long campaign responses sometimes ended before all JSON strings, braces, or brackets were complete. Writer instructions were tightened, content-length targets were reduced, and token settings were adjusted.

Reasoning text mixed with output

One tested model returned internal reasoning text before the campaign JSON. Cleaning logic was added, but the model still proved unsuitable for the required output size and schema.

Model token-rate limitation

The qwen/qwen3.6-27b model was tested for Writer output but its available output-token-per-minute limit was too low for the complete campaign. The Writer was returned to openai/gpt-oss-20b.

Structured Writer output

A Structured Output Parser was added only to the Writer, where complex nested output repeatedly failed. An auto-fix model repairs formatting when the first parsing attempt fails.

Object-versus-text parsing

After structured output was introduced, Writer results arrived as an object under the output field. Parse Draft initially converted that object to “[object Object].” The parsing logic was updated to accept structured objects directly while retaining text-based JSON as a fallback.

Provider rate limits

The Revision Agent failed when earlier model calls consumed too much of the account’s token-per-minute allowance. Its maximum token setting was reduced, and a 30-second delay was added before revision.

Misleading review label

The final campaign originally displayed pre-revision feedback under “Reviewer Summary,” which made it appear to describe the corrected campaign. The label was changed to “Pre-Revision Reviewer Feedback” to clarify that it triggered the automated revision.

These iterations improved functional coverage from 7 of 10 baseline tests to successful completion of all three targeted retests.

18. Key Learnings

This project demonstrated that multi-agent workflows require more than connecting several language-model nodes. Reliable operation depends on deterministic validation, clearly separated responsibilities, structured data transfer, parsing checks, controlled routing, and safe failure responses.

Prompt instructions alone do not guarantee valid JSON. Structured output is most valuable where the response is large and deeply nested, but it adds token usage, latency, and workflow complexity.

Low temperature improves consistency but does not eliminate empty responses, truncation, malformed output, or provider rate limits. Model capability and account-level token limits must therefore be considered part of the system design.

Automated review and automated revision serve different purposes. A score generated before revision should not be presented as a score for the revised output. Clear labeling and human approval remain essential.

Finally, preserving baseline failures and recording retests separately provides stronger evidence than overwriting unsuccessful results. It shows how evaluation directly informed the workflow improvements.

19. Limitations

The current prototype has several limitations:

- It supports only the three fictional products stored in the catalog.
- Product matching depends on identifiable product names rather than broader conversational understanding.
- Model-provider rate limits can cause intermittent empty responses, delays, or failed executions.
- Structured-output repair adds another model call, increasing latency and token usage.
- The workflow permits only one automated revision.
- Revised content is not sent through a second automated review.
- Reviewer scores describe the original draft and may not represent the corrected campaign.
- Optional URLs are captured but are not opened, researched, or verified.
- The workflow does not maintain campaign history, approvals, or version tracking in a separate database.
- The generated content is an initial draft and still requires human verification.
- No content is automatically published or distributed.

20. Future Improvements

Potential future improvements include:

- Add a post-revision review step that scores the corrected campaign without creating an unlimited loop.
- Store initial drafts, reviews, revisions, and final approvals for auditability.

- Add deterministic checks for required fields, word counts, and exactly three advertisement variations before calling the Reviewer.
- Introduce controlled retries for intermittent empty model responses.
- Use models or service tiers with higher and more predictable token limits.
- Add approval buttons and an explicit human decision record.
- Support additional catalog sources or a dedicated product-information database.
- Add role-based access and authentication for business users.
- Create a custom user interface for submitting requests and reviewing each content format.
- Add optional publishing integrations only after explicit human approval.
- Expand evaluation with more products, tones, adversarial inputs, and unsupported-claim tests.
- Track cost, token usage, latency, failure rate, and structured-repair frequency over time.

21. Tools and AI Assistance

The project used the following tools:

- n8n Cloud for low-code workflow orchestration, chat input, routing, parsing, error handling, and final response formatting.
- Google Sheets for the synthetic product catalog and evaluation workbook.
- Groq API for hosted language-model access.
- openai/gpt-oss-20b for strategy, writing, revision, and structured-output repair.
- openai/gpt-oss-120b for independent campaign review.
- Visual Studio Code for inspecting and sanitizing the exported workflow JSON.
- GitHub for version control and submission.
- Google Docs for project documentation.
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22. Implementation Evidence and Links

The following materials provide implementation and evaluation evidence:

GitHub repository: <https://github.com/DivyaReddyDodda/sprintpulse-gtm-agent>

Workflow demo: <https://www.loom.com/share/86fce9b0118c49e596a06c977f1e3b12>

Evaluation workbook:

https://docs.google.com/spreadsheets/d/1YGud7KfYXiBS_iJQMI0xEbnDLI9jKkWAcnEL_TKoml/edit?usp=sharing

Repository contents include:

- Sanitized n8n workflow JSON
- Synthetic product catalog
- Evaluation workbook with baseline tests and targeted retests
- Project documentation
- README with setup and usage instructions

Suggested implementation figures:

- Complete n8n workflow
- Input normalization and validation route
- Google Sheets product lookup
- Strategy and Writer Agent configuration
- Structured Writer output and parsing
- Reviewer PASS/REVISE decision
- Single-revision route
- Final pending-human-review response
- Evaluation summary

No API keys, passwords, personal email addresses, credential identifiers, or private Google Sheet document IDs are included in the repository.